

University Interscholastic League
2022 – 2023 Junior High Number Sense Test C

Contestant's Number _____

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Final		
2 nd		
1 st		
	Score	Initials

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.

Stop – Wait for Signal!

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|--|---|
| <p>(1) $39 + 282 =$ _____</p> <p>(2) $17 \times 6 =$ _____</p> <p>(3) $63 - 36 =$ _____</p> <p>(4) $105 \div 7 =$ _____</p> <p>(5) $\frac{9}{16} + \frac{3}{8} =$ _____</p> <p>(6) $48 \div 0.8 =$ _____</p> <p>(7) $28 \times 0.25 =$ _____</p> <p>(8) $14^2 =$ _____</p> <p>(9) $11 \times 5 \times 4 =$ _____</p> <p>*(10) $1976 + 1951 + 12 + 2023 =$ _____</p> <p>(11) $32 \times 1.5 =$ _____</p> <p>(12) The mode of 2, 3, 5, 2, 5 and 2 is _____</p> <p>(13) Which is larger: $\frac{8}{11}$ or 0.73? _____</p> <p>(14) $43 \times 47 =$ _____</p> <p>(15) $28 \times 12 =$ _____</p> <p>(16) $130 - 41 - 29 =$ _____</p> <p>(17) $83^2 =$ _____</p> <p>(18) $10 + 13 + 16 + 19 + 22 =$ _____</p> <p>(19) MMXXIV = _____ (Arabic Numeral)</p> | <p>*(20) $222.222\% \times 1809 =$ _____</p> <p>(21) $0.85 \times 18 - 0.35 \times 18 =$ _____</p> <p>(22) $761 \times 111 =$ _____</p> <p>(23) If $f(x) = 16 + 2x^2$, then $f(4) =$ _____</p> <p>(24) $8\frac{3}{4} \times 8\frac{1}{4} =$ _____ (mixed number)</p> <p>(25) $9\frac{7}{8} + 5\frac{3}{4} =$ _____</p> <p>(26) $39^2 + 39 =$ _____</p> <p>(27) The cube root of -27 is _____</p> <p>(28) If n is to 4 as 9 is to 12, then $n =$ _____</p> <p>(29) If $15 + 9x$ is 24 then $x =$ _____</p> <p>*(30) $2023 \times 5 + 2022 =$ _____</p> <p>(31) $106 \times 105 =$ _____</p> <p>(32) The sum of the two smallest prime numbers greater than 40 is _____</p> <p>(33) If $0.25 + 0.25 = n$, the $n^{-1} =$ _____</p> <p>(34) 47 is divisible by how many whole numbers? _____</p> <p>(35) One-eighth of a square mile = _____ acres</p> <p>(36) 6 percent = _____ (common fraction)</p> <p>(37) The total cost of item, that costs \$1, with a sales tax of $8\frac{1}{2}\%$ is \$ _____</p> |
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- (38) $101 \times 738 =$ _____
- (39) The area of an equilateral triangle with an altitude of 3-cm is $a\sqrt{3}$ and $a =$ _____ cm^2
- *(40) $250\sqrt{57600} =$ _____
- (41) 72 (base 8) = _____ (base 2)
- (42) $3^3 - 2^4 =$ _____
- (43) The GCF of 18, 24 and 12 is _____
- (44) What is the area of a trapezoid with base 15, 17 and height 12? _____
- (45) 33 (base 4) $\times 3$ (base 4) = _____ (base 4)
- (46) If $24 \div 3 = n \div 8$, then $n =$ _____
- (47) What is the length of an edge of a cube with surface area 24? _____
- (48) $3.5 \text{ m}^2 =$ _____ cm^2
- (49) If $15 - 5x < 55$, then $2x >$ _____
- *(50) $32^3 =$ _____
- (51) In the sequence: 1, 3, a , 7, 9, b , 13, . . .
 $2a + b^2 =$ _____
- (52) What is the 5th term in the sequence 32, 8, 2, . . ? _____
- (53) What is the sum of the interior angles for a convex polygon with 3 sides? _____ $^\circ$
- (54) What is the product of the LCM and GCD of 22 and 16? _____
- (55) What is the length of the side of a rectangle with perimeter 24 and width 5? _____
- (56) What is the radius of a circle with a circumference of 64π ? _____
- (57) If set $\mathbf{A} = \{1, 2, 3, \dots, 19\}$ and set $\mathbf{B} = \{5, 10, 15, \dots, 20\}$, then the number of elements in $\mathbf{A} \cap \mathbf{B}$ is _____
- (58) $(24^2 + 5 \times 9) \div 6$ has a remainder of _____
- (59) The number of vertices for a square pyramid is _____
- *(60) $96^2 \times 101 \div 16 =$ _____
- (61) $(2^{-3} + 2^{-2})^{-1} =$ _____
- (62) 660 yards = _____ mile
- (63) 36 feet/min = _____ yards/sec
- (64) 12% of 20 = _____
- (65) $10^7 \div 7$ has a remainder of _____
- (66) If the volume of a right circular cylinder, with radius 4 and length 12 is $k\pi$, what is k ? _____
- (67) The sum of the positive divisors of 41 is _____
- (68) $1! - 0! =$ _____
- (69) $9\frac{3}{4} \div \frac{3}{8} =$ _____
- *(70) $3000\pi^2 =$ _____
- (71) $28^2 + 4^2 =$ _____
- (72) $18\frac{2}{11}\% =$ _____ (common fraction)
- (73) $0.2666\dots =$ _____ (common fraction)
- (74) The volume of a right circular cone with height 9 and radius 6 is $k\pi$, and $k =$ _____
- (75) The sum of the 8th and 7th triangular numbers is _____
- (76) $16^2 \times 25 =$ _____
- (77) If the shortest distance between the points (0, 12) and (x , 0) is 13, what is x , if $x > 0$? _____
- (78) $\frac{6!}{3!} - 100 =$ _____
- (79) $\left(18\frac{1}{2}\right)^2 - \left(1\frac{1}{2}\right)^2 =$ _____
- *(80) 12 miles = _____ yards

2022 – 2023 University Interscholastic League Junior High Number Sense Test C – Key

(1) 321	*(20) 3819 – 4220	(38) 74538	(59) 5
(2) 102	(21) 9	(39) 3	*(60) 55268 – 61084
(3) 27	(22) 84471	*(40) 57000– 63000	(61) $\frac{8}{3}; 2\frac{2}{3}$
(4) 15	(23) 48	(41) 111010	(62) $\frac{3}{8}; .375$
(5) $\frac{15}{16}; .9375$	(24) $72\frac{3}{16}$	(42) 11	(63) $\frac{1}{5}; .2$
(6) 60	(25) $15\frac{5}{8}; \frac{125}{8}; 15.625$	(43) 6	(64) $2.4; 2\frac{2}{5}; \frac{12}{5}$
(7) 7	(26) 1560	(44) 192	(65) 3
(8) 196	(27) -3	(45) 231	(66) 192
(9) 220	(28) 3	(46) 64	(67) 42
*(10) 5664 – 6260	(29) 1	(47) 2	(68) 0
(11) 48	*(30) 11531 – 12743	(48) 35000	(69) 26
(12) 2	(31) 11130	(49) -16	*(70) 28129 – 31089
(13) .73	(32) 84	*(50) 31130 – 34406	(71) 800
(14) 2021	(33) 2	(51) 131	(72) $\frac{2}{11}$
(15) 336	(34) 2	(52) $\frac{1}{8}; .125$	(73) $\frac{4}{15}$
(16) 60	(35) 80	(53) 180	(74) 108
(17) 6889	(36) $\frac{3}{50}$	(54) 352	(75) 64
(18) 80	(37) 1.09	(55) 7	(76) 6400
(19) 2024		(56) 32	(77) 5
		(57) 3	(78) 20
		(58) 3	(79) 340
			*(80) 20064 – 22176

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.